ADITYA NARAYAN JHA

Applied AI Engineer
+91-9264906478 | Bangalore

work.aditya2025@gmail.com
linkedin.com/in/adityanj
github.com/HalfBloodPrince07

SUMMARY

Results-driven Applied AI Engineer with 3.5 years of experience taking complex LLM systems from concept to production. Specializes in multi-agent orchestration (LangGraph), advanced retrieval architectures (Vision-RAG, hybrid search), and heavy inference optimization. Proven track record of driving efficiency, leveraging TensorRT-LLM compilation and quantization to cut serving latency by 3x and significantly reduce GPU spend. An end-to-end owner proficient in Python, FastAPI, Docker, and AWS Bedrock, dedicated to maintaining high-quality user experiences through rigorous prompt engineering and continuous evaluation.

SKILLS

- **LLMs & Agents:** LangGraph, LangChain, AWS Strands, MCP, A2A message passing, Hugging Face Transformers, PyTorch, TensorFlow, Prompt Engineering (ToT, RAG, DSP), Ragas.
- **Retrieval & Search:** RAG, Vision-RAG, Hybrid Search (Vector + BM25), Cross-encoder Reranking, Embeddings, OpenSearch
- **Inference & Optimization:** TensorRT-LLM, Quantization, Model Compilation, GPU Serving Tuning, Ollama, LM Studio, Local Multi-GPU Deployment
- **Backend & Cloud**: Python, FastAPI, AWS (Bedrock, Lambda, S3), Docker, Linux, SQLite, SQLAlchemy, Pandas, NumPy
- **Frontend:** React, Vite, JavaScript/TypeScript, D3.js, Server-Sent Events (SSE), HTML5/CSS3

EXPERIENCE

Applied AI Engineer, Capgemini ***(May 2024 - Present)***

- **Engineered and deployed a LangGraph multi-agent framework** to automate end-to-end SDLC workflows, coordinating 5+ specialized agents across code generation, test authoring, and PR reviews to **reduce manual review time by 40%**.
- **Accelerated open-source LLM inference by 3x** (LLaMA, Phi-3) through **TensorRT-LLM compilation and model quantization**, maintaining near-zero accuracy degradation while driving significant cloud GPU cost reductions.
- **Architected and modernized RAG and Vision-RAG pipelines**, elevating document understanding accuracy by **28%** across complex, multimodal enterprise data inputs.
- **Developed an automated LLM test-generation Python microservice**, parsing Go ASTs to automatically emit executable Go test suites and drastically reduce manual QA overhead.
- **Delivered a multi-agent clinical decision support (CDS) solution** for a healthcare enterprise, accelerating turnaround times for critical clinician-facing recommendations.
- **Standardized agentic architecture** across 3 enterprise applications by integrating contextual reasoning and task-planning modules for multi-step workflow decomposition.
- **Established cross-domain prompt engineering and evaluation standards**, measurably reducing hallucination rates and increasing response reliability across 4 customer-facing applications.

GenAI Engineer, NEC India ***(January 2023 – May 2024)***

- **Spearheaded the architecture and development** of secure document generation and Business Document Connectivity (BDC) systems, scaling pipeline infrastructure to process **10,000+ enterprise documents monthly**.
- **Engineered advanced prompting techniques**—including Tree-of-Thought (ToT), Directional Stimulus, and RAG frameworks—with open-source LLMs (**Mistral-7B**, **Nous-Hermes-Llama-2-13B**), driving a **30% lift in response contextual relevance**.
- **Co-led a cross-functional team of 8 engineers** to seamlessly integrate LLM-driven features into enterprise applications, accelerating feature deployment cycles by 15%.

Selected Projects

Vantage ***(November 2025 - Present)***

Python, FastAPI, React, LangGraph, OpenSearch, Ollama, D3.js

- **Engineered a local-first semantic document search platform** powered by a 15+ agent LangGraph orchestration layer to handle dynamic intent classification, multi-step reasoning, and per-query retrieval strategies.
- **Built a hybrid retrieval architecture** combining 768-dimensional dense vector search with BM25 sparse keyword matching, utilizing cross-encoder reranking to maximize top-5 precision.
- **Architected a cloud-independent 5-tier memory system** (session, episodic, procedural, profile, agentic) to maintain cross-session context, surfaced via interactive D3.js knowledge graphs for entity exploration.
- **Optimized full-stack local inference**—including embeddings, cross-encoder reranking, and generation—to run on a single consumer GPU via Ollama with zero cloud API costs.

Podcast Production Pipeline ***(November 2025 - Present)***

Python, FastAPI, React/Vite, Qwen TTS, Hugging Face Transformers, LM Studio

- **Developed an autonomous multi-agent production system** that transforms topic prompts into audio podcasts using discrete cooperating agents for web research, scriptwriting, fact-checking, and sound design.
- **Integrated Qwen TTS via local Hugging Face Transformers**, enforcing dynamic pacing controls (140–170 WPM) and injecting custom prosody markers for natural, expressive audio delivery.
- **Implemented real-time progress streaming over Server-Sent Events (SSE)** to a React/Vite frontend, enabling automated rendering and export of MP3 audio, SRT subtitles, and timestamped transcripts.

EDUCATION

Bachelor of Technology | Information Technology ***(2019 – 2023)***

PSIT College of Engineering

ACHIEVEMENT

- Earned the **Capgemini Gem Award (2024)** for pioneering Agentic AI solutions, advancing multi-agent architectures and enterprise automation.